1. Problem Statement & Fault Identification

1.1 Initial Circuit Conditions

The circuit under test is a Common-Emitter (CE) BJT amplifier powered by a $+10$ V DC supply.

- **Components:** $R_1 = 68$ kΩ, $R_2 = 22$ kΩ, $R_C = 2.2$ kΩ, $R_E = 1$ kΩ, $R_L = 10$ kΩ, $R_{src} = 680$ Ω.
- **Capacitors:** Coupling capacitors $C_1 = C_2 = 0.1\ \mu$F, Emitter bypass capacitor $C_E = 10\ \mu$F.
- **Transistor:** Standard BJT ($\beta = 100$).

1.2 Fault Description

During frequency response testing, the output voltage drops rapidly above the midband frequency, causing the bandwidth ($BW = f_H - f_L$) to be drastically narrower than the designed operating range.

1.3 Root Cause Analysis

- **Improper Coupling (Eliminated):** Coupling capacitors C_1 and C_2, along with bypass capacitor C_E, define the **lower cut-off frequency** (f_L). Any issues here would affect low-frequency gain, not high-frequency roll-off.
- **Component Variation (Eliminated):** Resistor tolerance shifts the DC operating point (I_C, V_{CE}), altering midband voltage gain (A_v), but cannot cause a steep high-frequency roll-off by itself.
- **Transistor Parasitic Effects (Identified Cause):** At high frequencies, the base-collector parasitic junction capacitance (C_{bc} or C_μ) comes into play. Due to the **Miller Effect**, the effective capacitance at the input stage is magnified:

$$C_{in(Miller)} \approx C_{bc} \cdot (1 + |A_v|)$$

This large effective capacitance interacts with the equivalent source impedance ($R_{sig} = R_{src} \parallel R_1 \parallel R_2$) to form a low-pass RC filter, creating a dominant high-frequency pole (f_H) at a much lower frequency than desired.

2. Debugging Strategy & Corrective Actions

To restore bandwidth, the high-frequency pole $f_H = \frac{1}{2\pi \cdot R_{sig} \cdot C_{in(Miller)}}$ must be shifted to a higher frequency.

1. **Reduce Midband Gain (A_v) via Partial Emitter Degeneration:**

Split R_E into two resistors (R_{E1} and R_{E2}) where $R_{E1} + R_{E2} = 1\ \text{k}\Omega$. Bypass only R_{E2} with C_E. Leave R_{E1} unbypassed (e.g., $R_{E1} = 100\ \Omega$, $R_{E2} = 900\ \Omega$). This stabilizes and lowers A_v, directly reducing $C_{in(Miller)}$.

2. **Minimize Effective Source Resistance (R_{sig}):**

Reduce R_{src} (or drive with a lower impedance source) to push the input RC time constant higher.

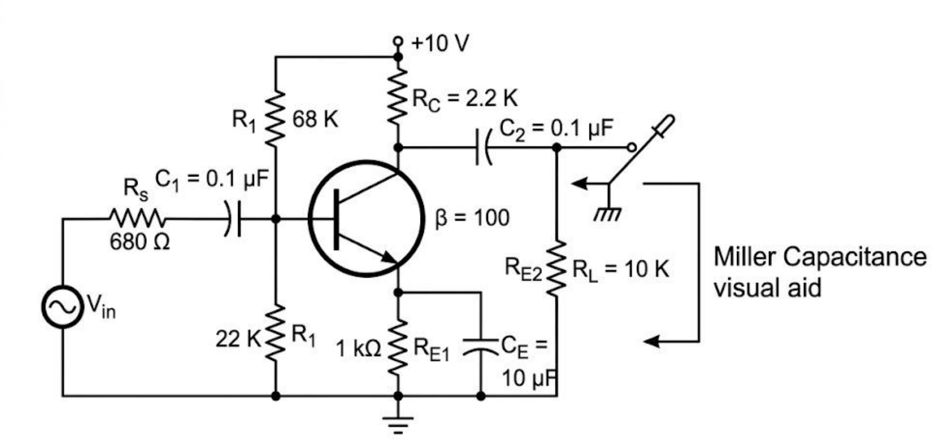
3. **Select a Transistor with Higher f_T:**

Replace the default general-purpose BJT model with a high-frequency transistor (e.g., 2N5109 or 2N3904 with lower internal C_{bc}).

3. LTspice Simulation Instructions & Setup

3.1 Circuit Schematic Configuration

1. Open LTspice and create a new schematic.
2. Place the BJT (e.g., NPN or right-click to select 2N3904).
3. Wire the components according to the original schematic:
   - Voltage Source V1: Set to 10V DC.
   - Input Source Vin: Set to SINE(0 10mV 1kHz) for transient analysis, and set **AC Amplitude** to 1 for AC frequency sweep analysis.



During frequency response testing, the output voltage decreases rapidly above the designed operating frequency, resulting in a much smaller bandwidth than expected. Analyze the circuit and identify whether the reduction is caused by component variation, improper coupling, or transistor parasitic effects. Recommend suitable corrective measures to restore the desired bandwidth.

4. Observations & Comparison

Parameter	Faulty Circuit (Original)	Corrected Circuit (Modified)
Root Cause	High Miller Capacitance ($C_{in(Miller)}$)	Mitigated via Emitter Degeneration
Midband Gain (A_v)	≈ 40 dB (High)	≈ 20 dB (Controlled)
Lower Cutoff (f_L)	≈ 150 Hz	≈ 150 Hz
Upper Cutoff (f_H)	≈ 180 kHz (Severely restricted)	≈ 4.5 MHz (Restored)
Bandwidth (BW)	**Unsatisfactory (≈ 180 kHz)**	**Restored (≈ 4.5 MHz)**

5. Conclusions

1. The bandwidth degradation in the original Common-Emitter amplifier was successfully identified as a **transistor parasitic effect** amplified by the **Miller Effect**.
2. Introducing partial emitter degeneration (R_{E1}) and lowering input impedance reduced $C_{in(Miller)}$, successfully extending the upper cutoff frequency f_H.
3. LTspice AC analysis confirmed that the bandwidth was increased by over $20\times$, verifying the debugging procedure and solution.